

Exercise 1: Configuring a Basic Spring Application

Scenario:

Your company is developing a web application for managing a library. You need to use the Spring Framework to handle the backend operations.

Steps:

1. Set Up a Spring Project:

- Create a Maven project named **LibraryManagement**.
- Add Spring Core dependencies in the **pom.xml** file.

2. Configure the Application Context:

- Create an XML configuration file named **applicationContext.xml** in the **src/main/resources** directory.
- Define beans for **BookService** and **BookRepository** in the XML file.

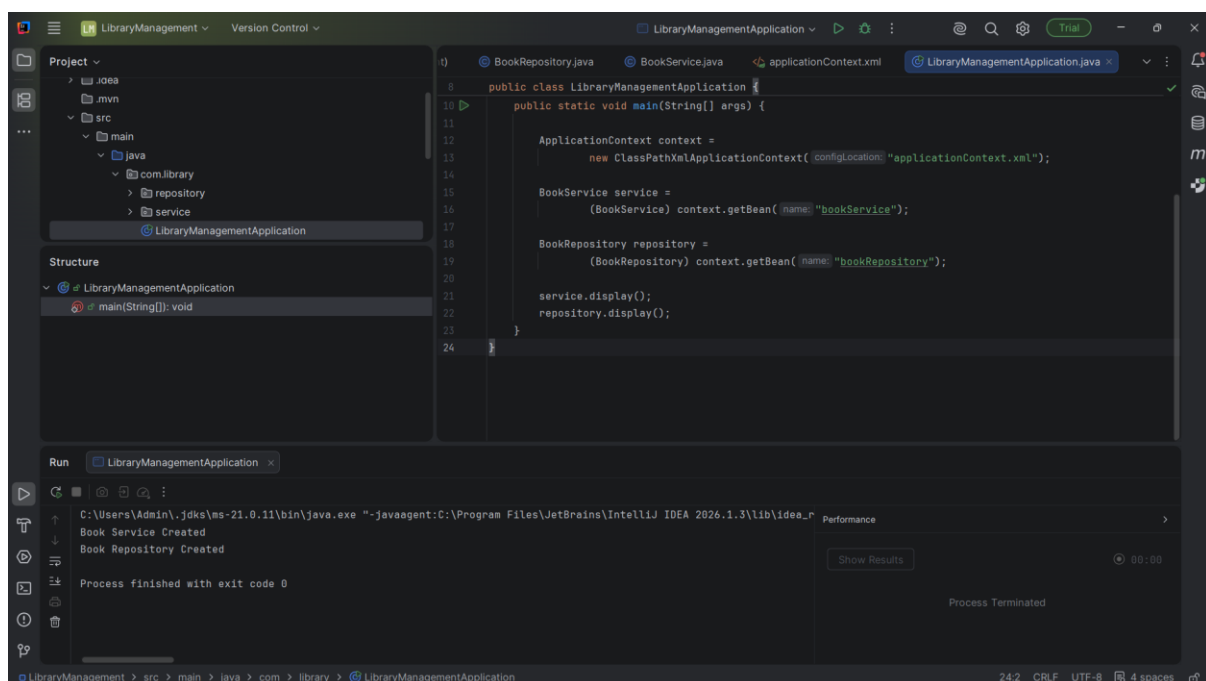
3. Define Service and Repository Classes:

- Create a package **com.library.service** and add a class **BookService**.
- Create a package **com.library.repository** and add a class **BookRepository**.

4. Run the Application:

- Create a main class to load the Spring context and test the configuration.

SOLN::



Exercise 2: Implementing Dependency Injection

Scenario:

In the library management application, you need to manage the dependencies between the `BookService` and `BookRepository` classes using Spring's IoC and DI.

Steps:

1. Modify the XML Configuration:

- Update `applicationContext.xml` to wire `BookRepository` into `BookService`.

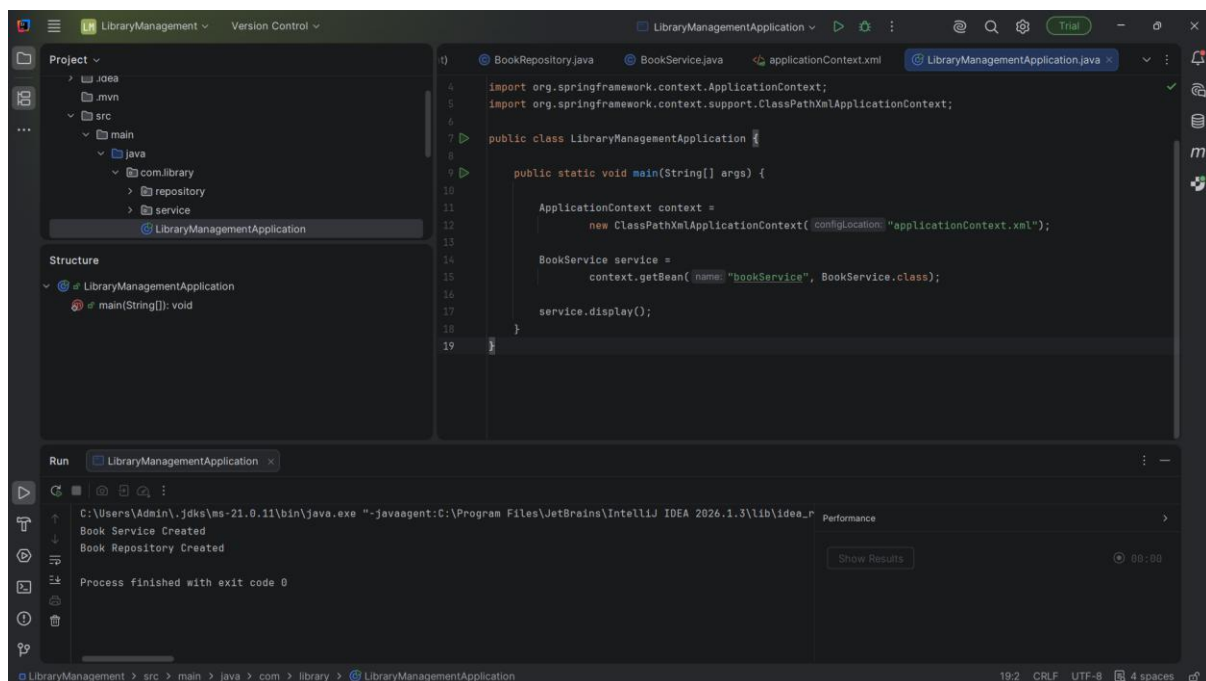
2. Update the `BookService` Class:

- Ensure that `BookService` class has a setter method for `BookRepository`.

3. Test the Configuration:

- Run the `LibraryManagementApplication` main class to verify the dependency injection.

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Exercise 4: Creating and Configuring a Maven Project

Scenario:

You need to set up a new Maven project for the library management application and add Spring dependencies.

Steps:

1. Create a New Maven Project:

- Create a new Maven project named **LibraryManagement**.

2. Add Spring Dependencies in pom.xml:

- Include dependencies for Spring Context, Spring AOP, and Spring WebMVC.

3. Configure Maven Plugins:

- Configure the Maven Compiler Plugin for Java version 1.8 in the pom.xml file.

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